

Robust Improvement Paths in Data Envelopment Analysis

Abstract

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1. Robust Improvement Paths in Data Envelopment Analysis

The attainment of an ambitious efficiency target in a single step may require changes that are too large to be regarded as operationally realistic. For this reason, it is useful to replace a one-shot recommendation with a sequence of intermediate states leading from the current position of the assessed unit to the final target. Such a sequence will be called a *path*. The construction of paths requires two ingredients. First, it is necessary to specify how candidate paths are generated. Second, once a set of candidates has been obtained, it is necessary to indicate how the most suitable one is selected. The former issue concerns the logic of the milestones that structure the transition toward the final recommendation. The latter concerns the criteria according to which different candidate paths are compared.

Let $\mathcal{D} = \{1, \dots, K\}$ be the set of decision making units. For each $k \in \mathcal{D}$, let

$$x_k = (x_{1k}, \dots, x_{Mk}) \in \mathbb{R}_+^M, \quad y_k = (y_{1k}, \dots, y_{Nk}) \in \mathbb{R}_+^N$$

denote, respectively, the vectors of inputs and outputs. Let $o \in \mathcal{D}$ be the unit under improvement and let

$$z^0 = (x_o, y_o)$$

be its current state. For every admissible state z , let

$$E_*(z), E^*(z), R^*(z), R_*(z)$$

denote, respectively, the worst and best efficiency scores, and the best and worst ranks. Moreover, let $\succeq_E^N$ denote the necessary efficiency relation.

A path leading from the current state of DMU_{*o*} is a finite sequence

$$\pi = (z^0, z^1, \dots, z^L),$$

where $L = L(\pi) \geq 1$ is the number of stages, $z^0 = (x_o, y_o)$, and z^L satisfies the final target. The state z^h is interpreted as the recommended situation after stage h , for each $h = 1, \dots, L$. In the standard improvement setting,

$$x^h = x^{h-1} - b_x^h, \quad y^h = y^{h-1} + b_y^h,$$

where $b_x^h \geq 0$ and $b_y^h \geq 0$ are interpreted as input reductions and output increases carried out between two consecutive stages.

It is useful to distinguish three kinds of paths. A path is called *real* if every intermediate state z^h , $h = 1, \dots, L$, coincides with the performance vector of an existing unit. A path is called *fictive* if at least one intermediate state is generated by optimization and does not coincide with an existing unit. Finally, a path is called *mixed* if both types of states occur along the transition. This distinction is methodologically important. Real paths are easier to interpret because every step can be associated with an observable comparator. Fictive paths are more flexible because they

are not constrained by the empirical distribution of units and, therefore, may reflect intermediate goals that are not explicitly represented in the data. Mixed paths combine both advantages.

1.1. Generation of candidate paths

The generation of candidate paths may be based on several robust objects derived from the model. In what follows, five families of generators are considered. The first two exploit the robust ordinal structure induced by the necessary efficiency relation. The next two distribute intermediate milestones over intervals of best ranks and best efficiency scores. The fifth regulates the width of the robustness interval itself. In each case, the same general idea applies: once the milestones are fixed, a path is obtained either by selecting existing units that satisfy or approximate these milestones, or by generating fictive intermediate states by means of multi-objective optimization.

1.2. Paths generated from the Hasse diagram of the necessary relation

The necessary efficiency relation provides the strongest robust pairwise information available in the model. If a unit is necessarily at least as efficient as another one, this statement holds for all admissible combinations of weights. Consequently, the Hasse diagram of this relation yields a natural skeleton for multi-stage improvement: each move along the diagram leads to a state that is robustly better than the previous one.

Since the necessary relation may contain mutual ties, it is convenient to quotient the set of units by the corresponding equivalence. For any $k, \ell \in \mathcal{D}$, let

$$k \sim_N \ell \iff (k \succeq_E^N \ell \text{ and } \ell \succeq_E^N k).$$

Let $[k]$ denote the equivalence class of k , and let

$$\overline{\mathcal{D}} = \mathcal{D} / \sim_N$$

be the quotient set. On $\overline{\mathcal{D}}$, let

$$[k] \prec_N [\ell] \iff \ell \succeq_E^N k \text{ and } \neg(k \succeq_E^N \ell).$$

Let $\prec_N$ denote the cover relation in the Hasse diagram of $(\overline{\mathcal{D}}, \prec_N)$.

A real path generated from the Hasse diagram is any sequence

$$\pi = (k_0, k_1, \dots, k_L)$$

such that

$$k_0 = o, \quad [k_{h-1}] \prec_N [k_h] \quad \text{for } h = 1, \dots, L, \quad [k_L] \in \max(\overline{\mathcal{D}}).$$

The associated state path is given by $z^h = (x_{k_h}, y_{k_h})$. In other words, a real Hasse path follows a chain of cover relations from the class of the assessed unit to one of the maximal classes.

The same ordinal logic may also be used to generate fictive intermediate states. To this end, let

$$[o] = C_0 \prec_N C_1 \prec_N \dots \prec_N C_L, \quad C_L \in \max(\overline{\mathcal{D}}),$$

be a chain of classes in the Hasse diagram. Then, for each stage $h = 1, \dots, L$, the milestone is defined by class C_h , while the actual state z^h is selected from the set of efficient attainable states satisfying the robust requirement associated with C_h . The advantage of the fictive version is that it preserves the strongest robust ordinal structure while allowing one to smooth the transition between consecutive stages.

The Hasse-based construction is attractive because it preserves the finest robust structure induced by the necessary relation. The real version emphasizes interpretability, since each stage is represented by an existing comparator. The

fictive version emphasizes flexibility, since it allows the recommended progression to be adjusted to managerial or technical requirements without abandoning the robust ordering logic.

1.3. Paths generated from efficiency fronts

The Hasse diagram provides a fine-grained view of the robust order. A coarser and often more manageable alternative is obtained by partitioning the quotient set into fronts. Each front contains classes that are maximal after removing all classes assigned to previous fronts. Such a decomposition yields a layered description of the robust efficiency structure and naturally suggests a stepwise progression from lower to higher layers.

Let

$$\mathcal{F}_1 = \max(\overline{\mathcal{D}}),$$

and, recursively, for $q = 2, \dots, Q$,

$$\mathcal{F}_q = \max\left(\overline{\mathcal{D}} \setminus \bigcup_{r=1}^{q-1} \mathcal{F}_r\right),$$

where Q is the total number of fronts. Let $q(o)$ be the index such that $[o] \in \mathcal{F}_{q(o)}$.

A real path generated from efficiency fronts is any sequence

$$\pi = (k_0, k_1, \dots, k_{q(o)-1})$$

such that

$$k_0 = o, \quad [k_h] \in \mathcal{F}_{q(o)-h} \quad \text{for } h = 0, \dots, q(o) - 1.$$

If a stronger form of monotonicity is desired, one may additionally require

$$[k_{h-1}] \prec_N [k_h] \quad \text{for } h = 1, \dots, q(o) - 1.$$

Hence, the path proceeds front by front until the first front is reached.

The same layered logic may be transferred to fictive states. In that case, the milestone at stage h is the front $\mathcal{F}_{q(o)-h}$ itself, whereas the actual intermediate state is selected from the set of efficient attainable states satisfying the robust requirements associated with this front. The resulting path preserves the coarse robust stratification of the units while allowing one to choose smoother or less costly intermediate states than those represented in the data.

This generator replaces the detailed local structure of the Hasse diagram with a sequence of increasingly better layers. Consequently, it is easier to control the number of stages and to communicate the managerial meaning of the progression. At the same time, it retains the robust character of the necessary relation, because the fronts are constructed from its quotient order.

1.4. Paths generated from equally distributed best ranks

A third possibility is to describe the desired progression by equally spaced milestones in terms of the best attainable rank. Such a construction is natural when the final recommendation is expressed in the language of rank improvement.

Let r^* be the final best-rank target, with

$$r^* \leq R^*(z^0) = R_o^*.$$

Fix a number of stages $L \geq 1$ and define the milestone sequence

$$r^h = \left\lceil R_o^* - \frac{h}{L}(R_o^* - r^*) \right\rceil, \quad h = 0, \dots, L.$$

Then

$$r^0 = R_o^*, \quad r^L = r^*.$$

Thus, the sequence $\{r^h\}_{h=0}^L$ divides the interval between the current best rank and the target best rank into approximately equal parts.

In the fictive version, a path generated from equally distributed best ranks is any state path

$$\pi = (z^0, z^1, \dots, z^L)$$

such that

$$R^*(z^h) \leq r^h \quad \text{for } h = 1, \dots, L,$$

and z^L satisfies the final target. This version ensures exact compliance with the prescribed milestone structure.

In the real version, exact milestone attainment may be impossible because the dataset may not contain units located exactly at the prescribed milestone ranks. In that case, the nearest represented levels are selected. More precisely, for each stage $h = 1, \dots, L$, let

$$\varepsilon_h^R = \min_{k \in \mathcal{D}: R_k^* \leq r^{h-1}} |R_k^* - r^h|,$$

and define

$$\mathcal{D}_h^{R, \text{real}} = \{k \in \mathcal{D} : R_k^* \leq r^{h-1}, |R_k^* - r^h| = \varepsilon_h^R\}.$$

A real best-rank path is then any sequence

$$\pi = (k_0, k_1, \dots, k_L)$$

such that $k_0 = o$ and

$$k_h \in \mathcal{D}_h^{R, \text{real}} \quad \text{for } h = 1, \dots, L.$$

The fictive version preserves the regularity of the intended improvement most faithfully, while the real version provides a more directly interpretable sequence of observable role models. Both viewpoints are meaningful and complement one another.

1.5. Paths generated from equally distributed best efficiency scores

When the final recommendation is expressed through a target value of the best efficiency score, it is natural to distribute the milestones directly over the interval between the current and target values of $E^*(\cdot)$.

Let e^* be the final best-score target, with

$$e^* \geq E^*(z^0) = E_o^*.$$

Fix a number of stages $L \geq 1$ and define

$$e^h = E_o^* + \frac{h}{L}(e^* - E_o^*), \quad h = 0, \dots, L.$$

Then

$$e^0 = E_o^*, \quad e^L = e^*.$$

In the fictive version, a path generated from equally distributed best efficiency scores is any state path

$$\pi = (z^0, z^1, \dots, z^L)$$

such that

$$E^*(z^h) \geq e^h \quad \text{for } h = 1, \dots, L,$$

and z^L satisfies the final target.

In the real version, the dataset may again fail to contain units whose best efficiency scores coincide exactly with the milestones. Therefore, for each stage $h = 1, \dots, L$, let

$$\varepsilon_h^E = \min_{k \in \mathcal{D}: E_k^* \geq e^{h-1}} |E_k^* - e^h|,$$

and define

$$\mathcal{D}_h^{E, \text{real}} = \{k \in \mathcal{D} : E_k^* \geq e^{h-1}, |E_k^* - e^h| = \varepsilon_h^E\}.$$

A real best-score path is then any sequence

$$\pi = (k_0, k_1, \dots, k_L)$$

such that $k_0 = o$ and

$$k_h \in \mathcal{D}_h^{E, \text{real}} \quad \text{for } h = 1, \dots, L.$$

This generator is the score-based analogue of the best-rank construction. It is appropriate whenever the interpretation of the recommendation is naturally expressed in terms of attainable score levels rather than ordinal positions in the population.

1.6. Paths generated from equally distributed robustness widths

The previous two generators regulate the level of the attained efficiency. A complementary idea is to regulate the width of the robustness interval itself. This is useful when one wishes not only to improve the efficiency level, but also to reduce the sensitivity of the assessment to admissible weights.

Let the robustness-width indicator be defined by

$$W(z) = \begin{cases} E^*(z) - E_*(z), & \text{in the score-based setting,} \\ R_*(z) - R^*(z), & \text{in the rank-based setting.} \end{cases}$$

Let

$$W_0 = W(z^0).$$

Fix a desired final width w^* , with

$$0 \leq w^* \leq W_0,$$

and define the milestone sequence

$$w^h = W_0 - \frac{h}{L}(W_0 - w^*), \quad h = 0, \dots, L.$$

In the fictive version, a path generated from equally distributed robustness widths is any state path

$$\pi = (z^0, z^1, \dots, z^L)$$

such that

$$W(z^h) \leq w^h \quad \text{for } h = 1, \dots, L,$$

$$q(z^h) \geq q(z^{h-1}) \quad \text{for } h = 1, \dots, L,$$

for a chosen primary progress indicator q , and z^L satisfies the final target. The additional monotonicity constraint imposed on q is important. Width reduction alone does not guarantee progress toward the final recommendation. Therefore, the width milestones should be combined with a non-deterioration condition relative to the main target-oriented indicator.

A real version may be defined analogously by replacing each width milestone with the closest represented level in the data. In practice, however, the fictive version is the natural one. Indeed, intermediate and final width milestones may easily fail to be represented by existing units, especially after imposing the requirement of monotone progress toward the principal target. For this reason, width-based generation is especially well suited to optimization-generated intermediate states.

1.7. Stagewise budget constraints

The generators introduced above define how milestones are chosen. Independently of that choice, one may constrain the magnitude of the modifications allowed at each stage. Such a restriction is particularly natural when the very motivation for multi-stage planning is to avoid excessive adjustments in a single step.

For each factor i and each stage h , let $\bar{\delta}_{i,h} \geq 0$ be the maximum admissible normalized change. Alternatively, let $\bar{c}_h \geq 0$ be the maximum admissible aggregate effort at stage h .

For any generator $g \in \{H, F, R, E, W\}$, let $\Pi^g(o, T^*)$ be the corresponding set of candidate paths. Then the set of budget-capped candidate paths is

$$\Pi_B^g(o, T^*) = \{\pi \in \Pi^g(o, T^*) : \delta_{i,h}(\pi) \leq \bar{\delta}_{i,h} \ \forall i, h\},$$

or, in the aggregate version,

$$\Pi_B^g(o, T^*) = \{\pi \in \Pi^g(o, T^*) : c_h(\pi) \leq \bar{c}_h \ \forall h\}.$$

Thus, stagewise budgets are not a separate path generator. Rather, they constitute an overlay that can be imposed on any of the basic generators described above.

2. Measures for evaluating candidate paths

Once candidate paths have been generated, the next task is to identify the most suitable one. This requires measures capable of capturing distinct and sometimes competing aspects of path quality. Some measures evaluate the total or maximal effort required to follow a path. Others assess the smoothness of the modifications, the regularity of progress toward the target, the robustness of intermediate states, or the similarity of consecutive benchmarks. The same family of measures may be applied regardless of the generator used in the first stage. This is important, because it preserves a clean separation between the generation of candidates and their subsequent evaluation.

2.1. Normalized stagewise modifications

For each path $\pi = (z^0, \dots, z^L)$, each input $m = 1, \dots, M$, and each stage $h = 1, \dots, L$, let

$$\delta_{m,h}^x(\pi) = \frac{x_m^{h-1} - x_m^h}{r_m^x},$$

where $r_m^x > 0$ is a normalization constant, e.g., the observed range or the admissible range of modifications for input m or the range of observed value for the existing units. Similarly, for each output $n = 1, \dots, N$, let

$$\delta_{n,h}^y(\pi) = \frac{y_n^h - y_n^{h-1}}{r_n^y},$$

where $r_n^y > 0$ is the corresponding normalization constant for output n .

Let

$$\mathcal{J} = \{x_1, \dots, x_M, y_1, \dots, y_N\}$$

be the set of all adjustable factors, and let $\delta_{i,h}(\pi)$ denote the normalized change of factor $i \in \mathcal{J}$ at stage h . Define the total normalized change of factor i along the whole path by

$$\Delta_i(\pi) = \sum_{h=1}^{L(\pi)} \delta_{i,h}(\pi).$$

Let

$$\mathcal{J}^+(\pi) = \{i \in \mathcal{J} : \Delta_i(\pi) > 0\},$$

and, for each $i \in \mathcal{J}^+(\pi)$, let

$$\mu_i(\pi) = \frac{\Delta_i(\pi)}{L(\pi)}$$

be the average stagewise change of factor i .

Finally, let

$$c_h(\pi) = \sum_{i \in \mathcal{J}} \omega_i \delta_{i,h}(\pi),$$

where $\omega_i \geq 0$ and $\sum_{i \in \mathcal{J}} \omega_i = 1$, be the aggregate effort required at stage h .

2.2. Measures of total effort and worst-stage effort

The first family of measures assesses the global magnitude of the modifications required by the path. The total path effort is defined by

$$TC(\pi) = \sum_{h=1}^{L(\pi)} c_h(\pi).$$

The smaller the value of $TC(\pi)$, the smaller the cumulative effort needed to complete the whole transition.

A complementary measure is the maximum stage effort,

$$MSC(\pi) = \max_{h=1, \dots, L(\pi)} c_h(\pi).$$

This measure captures the hardest single step of the path. It is particularly relevant when abrupt one-stage changes are considered undesirable or infeasible.

If C^{dir} denotes the minimum one-shot effort needed to attain the final target, then the relative detour caused by gradual improvement may be measured by

$$DR(\pi) = \frac{TC(\pi)}{C^{\text{dir}}}.$$

Values of $DR(\pi)$ close to one indicate that the gradual path does not entail a substantial increase in total effort relative to the direct solution.

2.3. Measure of deviation from prescribed milestones

For Hasse- and front-based paths, milestone compliance is usually built into the path generator. By contrast, for real paths based on best ranks, best scores, or robustness widths, exact milestone attainment may be impossible because the corresponding levels are not represented in the data. For this reason, it is useful to measure how closely the path follows the prescribed sequence of milestones.

Let $g \in \{H, F, R, E, W\}$ denote the generator used to construct the path and let m_h^g denote the milestone prescribed at stage h . The deviation from prescribed milestones is defined by

$$MD_g(\pi) = \frac{1}{L(\pi)} \sum_{h=1}^{L(\pi)} d_h^g(\pi),$$

where

$$d_h^R(\pi) = |R^*(z^h) - r^h|, \quad d_h^E(\pi) = |E^*(z^h) - e^h|, \quad d_h^W(\pi) = |W(z^h) - w^h|.$$

For Hasse and front paths, one may set

$$d_h^H(\pi) = \begin{cases} 0, & \text{if the prescribed Hasse milestone is attained at stage } h, \\ +\infty, & \text{otherwise,} \end{cases}$$

and

$$d_h^F(\pi) = \begin{cases} 0, & \text{if the prescribed front milestone is attained at stage } h, \\ +\infty, & \text{otherwise.} \end{cases}$$

This measure is essential whenever the generator prescribes an ideal sequence of milestones but the actual path can only approximate it. In this sense, it naturally subsumes the idea of selecting the nearest represented milestones. Such a rule is best regarded not as a separate generator, but as a criterion used to compare already generated candidates.

2.4. Measures of balance of performance modifications over stages

A central rationale of multi-stage improvement is that the required modifications should be distributed in a balanced manner over consecutive stages. This idea may be captured in several complementary ways.

The first measure evaluates, for each factor that changes, the extent to which the total required modification is split evenly over all stages. It is defined by

$$BP(\pi) = \frac{1}{|\mathcal{J}^+(\pi)| L(\pi)} \sum_{i \in \mathcal{J}^+(\pi)} \sum_{h=1}^{L(\pi)} \left(\frac{\delta_{i,h}(\pi) - \mu_i(\pi)}{\Delta_i(\pi)} \right)^2.$$

Small values of $BP(\pi)$ indicate that the path distributes the required changes relatively evenly across the stages.

If balancing changes in some factors is more important than balancing changes in others, one may use the weighted version

$$WBP(\pi) = \frac{1}{L(\pi)} \sum_{i \in \mathcal{J}^+(\pi)} w_i^P \sum_{h=1}^{L(\pi)} \left(\frac{\delta_{i,h}(\pi) - \mu_i(\pi)}{\Delta_i(\pi)} \right)^2,$$

where $w_i^P \geq 0$ and

$$\sum_{i \in \mathcal{J}^+(\pi)} w_i^P = 1.$$

If some stages are regarded as more sensitive than others, for instance because it is particularly desirable to avoid large changes near the beginning or near the end of the process, one may define

$$SBP(\pi) = \frac{1}{|\mathcal{J}^+(\pi)|} \sum_{i \in \mathcal{J}^+(\pi)} \sum_{h=1}^{L(\pi)} v_h^P \left(\frac{\delta_{i,h}(\pi) - \mu_i(\pi)}{\Delta_i(\pi)} \right)^2,$$

where $v_h^P \geq 0$ and

$$\sum_{h=1}^{L(\pi)} v_h^P = 1.$$

Combining both types of weights yields

$$SWBP(\pi) = \sum_{i \in \mathcal{J}^+(\pi)} w_i^P \sum_{h=1}^{L(\pi)} v_h^P \left(\frac{\delta_{i,h}(\pi) - \mu_i(\pi)}{\Delta_i(\pi)} \right)^2.$$

A bottleneck-type indicator is given by

$$MCP(\pi) = \max_{i \in \mathcal{J}^+(\pi), h=1, \dots, L(\pi)} \frac{\delta_{i,h}(\pi)}{\Delta_i(\pi)}.$$

Whereas the previous measures evaluate dispersion around an average, $MCP(\pi)$ focuses on the largest single share of the total adjustment of a given factor. It is therefore especially informative when one wishes to avoid paths in which a specific factor absorbs most of the burden at one stage.

2.5. Measures of regularity of progress toward the final target

A path should not only be smooth in terms of modifications; it should also ensure regular advancement toward the final goal. A sequence in which some stages require substantial effort but yield very little progress may be considered unsatisfactory.

Let $q(z)$ be a progress indicator consistent with the final target. Depending on the nature of the target, one may set

$$q(z) = E^*(z), \quad q(z) = -R^*(z), \quad q(z) = E_*(z), \quad q(z) = -R_*(z).$$

For each stage $h = 1, \dots, L(\pi)$, let

$$\eta_h(\pi) = \frac{q(z^h) - q(z^{h-1})}{c_h(\pi)}$$

be the gain in target-oriented progress obtained per unit of stage effort.

The regularity of these gains may be evaluated by

$$PYV(\pi) = \frac{1}{L(\pi)} \sum_{h=1}^{L(\pi)} (\eta_h(\pi) - \bar{\eta}(\pi))^2,$$

where

$$\bar{\eta}(\pi) = \frac{1}{L(\pi)} \sum_{h=1}^{L(\pi)} \eta_h(\pi).$$

Small values of $PYV(\pi)$ indicate that the path offers a relatively regular gain per unit of effort over successive stages.

A complementary measure is

$$PYM(\pi) = \min_{h=1, \dots, L(\pi)} \eta_h(\pi).$$

Large values of $PYM(\pi)$ mean that no stage is particularly inefficient in terms of progress per unit of effort.

2.6. Measures of robustness width along the path

Even if robustness width is not used as the generating principle of the path, it remains an informative descriptor of path quality. A path may improve the target value while still passing through states whose assessments remain highly sensitive to admissible weights. For this reason, width-based measures are useful both when robustness width is itself a milestone variable and when it only plays the role of an auxiliary criterion.

The average robustness width along the path is given by

$$APW(\pi) = \frac{1}{L(\pi) + 1} \sum_{h=0}^{L(\pi)} W(z^h).$$

The final robustness width is

$$FW(\pi) = W(z^{L(\pi)}).$$

If robustness is considered especially important at some stages, one may also use

$$WW(\pi) = \sum_{h=0}^{L(\pi)} \lambda_h W(z^h),$$

where $\lambda_h \geq 0$ and

$$\sum_{h=0}^{L(\pi)} \lambda_h = 1.$$

These measures are particularly valuable because they remain meaningful independently of the path generator. Even if the path is generated from ranks, scores, fronts, or the Hasse diagram, narrow robustness intervals still indicate that the recommendations are less sensitive to the admissible weight space.

2.7. Measures of continuity of reference sets and preservation of the operational profile

When a path is interpreted as a sequence of benchmarking recommendations, it is natural to prefer consecutive stages that rely on relatively similar peer sets. Abrupt changes in the identity of the benchmarks may hinder interpretability and reduce managerial acceptability.

Let $P(z^h)$ denote the peer or reference set associated with state z^h . The continuity of peer sets may be evaluated by

$$PC(\pi) = 1 - \frac{1}{L(\pi)} \sum_{h=1}^{L(\pi)} \frac{|P(z^{h-1}) \cap P(z^h)|}{|P(z^{h-1}) \cup P(z^h)|}.$$

Small values of $PC(\pi)$ indicate that successive steps are benchmarked against relatively stable sets of peers.

A second consideration concerns the preservation of the operational profile of the unit. In some applications, an improvement path is more convincing if it does not fundamentally change the structural proportions of the activities performed by the unit. Let

$$s_m^x(z) = \frac{x_m}{\sum_{r=1}^M x_r}, \quad s_n^y(z) = \frac{y_n}{\sum_{t=1}^N y_t},$$

and let

$$s(z) = (s_1^x(z), \dots, s_M^x(z), s_1^y(z), \dots, s_N^y(z)).$$

Then the preservation of the operational profile may be measured by

$$OPP(\pi) = \sum_{h=1}^{L(\pi)} \|s(z^h) - s(z^0)\|_1.$$

Small values of $OPP(\pi)$ indicate that the path remains closer to the original composition of inputs and outputs.

2.8. Measure of realness

When both real and fictive candidates are considered, it may be useful to quantify the extent to which a path is supported by existing benchmark units. This may be done by means of

$$RR(\pi) = \frac{1}{L(\pi)} \sum_{h=1}^{L(\pi)} \mathbf{1}\{z^h \text{ coincides with an existing unit}\}.$$

Large values of $RR(\pi)$ mean that a larger share of the path is directly interpretable in terms of observable comparators.

3. Selection of the most suitable path from the pre-generated candidates

Let

$$\Pi_{\text{cand}}(o, T^*) = \bigcup_{g \in \{H, F, R, E, W\}} \Pi_B^g(o, T^*)$$

be the set of all pre-generated candidate paths satisfying the stagewise budget restrictions. The selection problem consists in identifying the path that best reconciles the desired properties. Since path generation and path evaluation play different roles, it is useful to maintain a clear separation between them. The first stage determines *what kinds of transitions are admissible*. The second stage determines *which admissible transition is the most appropriate*.

The simplest approach is based on a single measure. If M is any path-quality measure, then one may select

$$\pi^* \in \arg \min_{\pi \in \Pi_{\text{cand}}(o, T^*)} M(\pi)$$

for cost-type measures, or

$$\pi^* \in \arg \max_{\pi \in \Pi_{\text{cand}}(o, T^*)} M(\pi)$$

for benefit-type measures. This rule is appropriate when one property of the path is clearly dominant, for instance when the objective is to minimize the total effort, minimize the largest stage effort, or minimize the deviation from the prescribed milestones.

In many applications, however, no single measure is sufficient. It is then natural to combine several criteria in a hierarchical manner. Let $M_1, \dots, M_s$ be path-quality measures ordered by importance. Then one may select

$$\pi^* \in \text{lexmin}_{\pi \in \Pi_{\text{cand}}(o, T^*)} (M_1(\pi), \dots, M_s(\pi)).$$

A lexicographic rule is particularly natural in the present setting. When paths are generated by equally distributed ranks, scores, or widths, it is reasonable to first minimize the deviation from the prescribed milestones, then minimize the largest stage effort, and only afterward refine the choice by means of smoothness or robustness criteria. Thus, one may consider

$$\text{lexmin}(MD_g, MSC, SWBP, FW).$$

By contrast, for Hasse- and front-based paths, milestone attainment is usually structural, so selection may start directly from effort and smoothness measures, for instance through

$$\text{lexmin}(MSC, TC, SWBP, FW).$$

A third possibility is to retain only those candidates that are nondominated with respect to a chosen family of measures. Let $\mathcal{M}$ be a finite family of path-quality measures, all interpreted in the cost direction. A path $\pi \in \Pi_{\text{cand}}(o, T^*)$ is Pareto-efficient if there does not exist another path $\pi' \in \Pi_{\text{cand}}(o, T^*)$ such that

$$M(\pi') \leq M(\pi) \quad \text{for all } M \in \mathcal{M},$$

with at least one strict inequality. The Pareto approach is useful when several measures are simultaneously important and it is undesirable to impose a complete ordering among them at the outset. In such a case, the analyst may first compute the Pareto-efficient subset and then apply an auxiliary preference model or an interactive choice procedure.

Finally, one may normalize a selected family of measures and aggregate them into a single value function. If $\widetilde{M}_1, \dots, \widetilde{M}_s$ are normalized cost-type measures and $\alpha_1, \dots, \alpha_s$ are nonnegative weights such that

$$\sum_{r=1}^s \alpha_r = 1,$$

then one may minimize

$$V(\pi) = \sum_{r=1}^s \alpha_r \widetilde{M}_r(\pi).$$

This approach is appropriate when the decision maker is able to express trade-offs between different properties of the path.

The discussion above also clarifies the role of ideas such as choosing the nearest represented milestones or choosing the least costly path. These are not separate generators. Rather, they are selection principles applied to a pre-generated candidate set. The former corresponds to minimizing the deviation from prescribed milestones, whereas the latter corresponds to minimizing the total or maximal stage effort. Maintaining this distinction makes the overall methodology more transparent: candidate paths are generated first according to a specified structural logic, and are only afterward filtered or ranked according to the desired quality criteria.

References